# Experiment 14:- Write a Program to Implement K- Means Algorithm of Machine Learning on any Dataset.

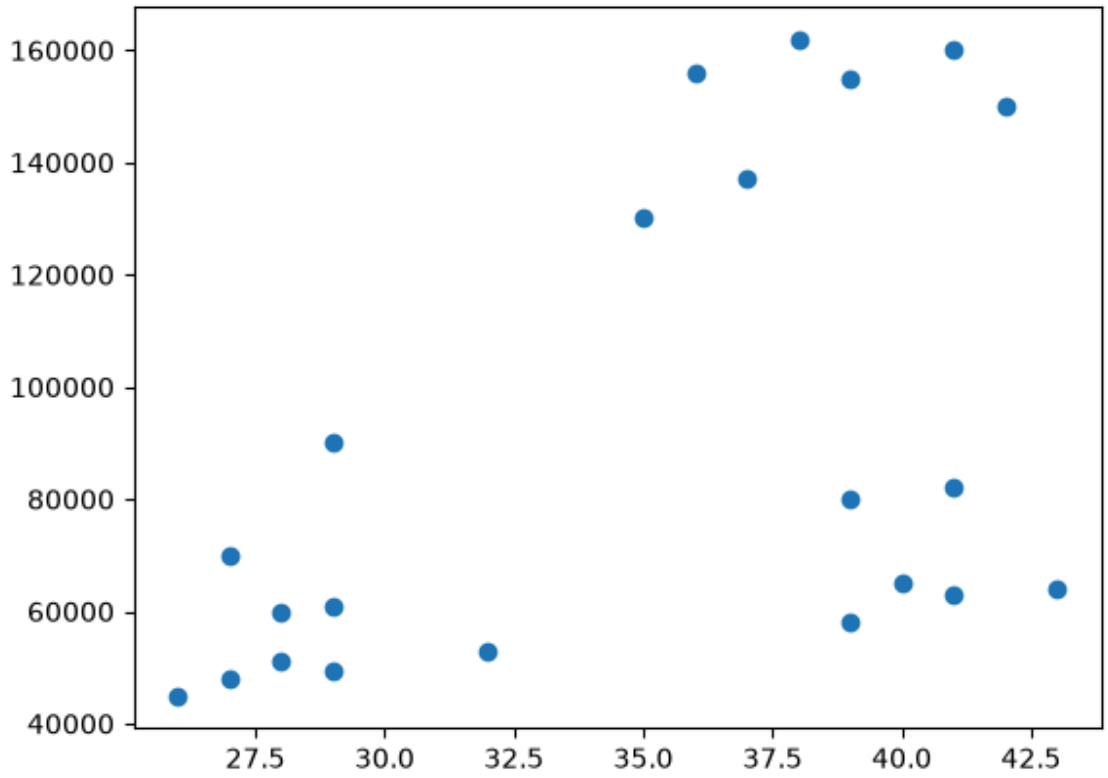
In [1]:
```
from sklearn.cluster import KMeans
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
import matplotlib.pyplot as plt
%matplotlib inline
```

In [2]:
```
df = pd.read_csv('kmeansdata.csv')
df.head()
```

Out[2]:

| | Name | Age | Income |
|---|---|---|---|
| **0** | Ram | 27 | 70000 |
| **1** | Shyam | 29 | 90000 |
| **2** | Manish | 29 | 61000 |
| **3** | Arjun | 28 | 60000 |
| **4** | Abhishek | 42 | 150000 |

In [3]:
```
plt.scatter(df.Age, df.Income)
plt.show()
```




In [4]:
```
km=KMeans(n_clusters=3)
km
```

Out[4]:

KMeans ⓘ ?

▶ Parameters

In [5]:
```
y_predicted = km.fit_predict(df[['Age','Income']])
y_predicted
```

Out[5]:
```
array([2, 2, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 0],
      dtype=int32)
```

In [6]:
```
df['cluster']= y_predicted
df.head()
```

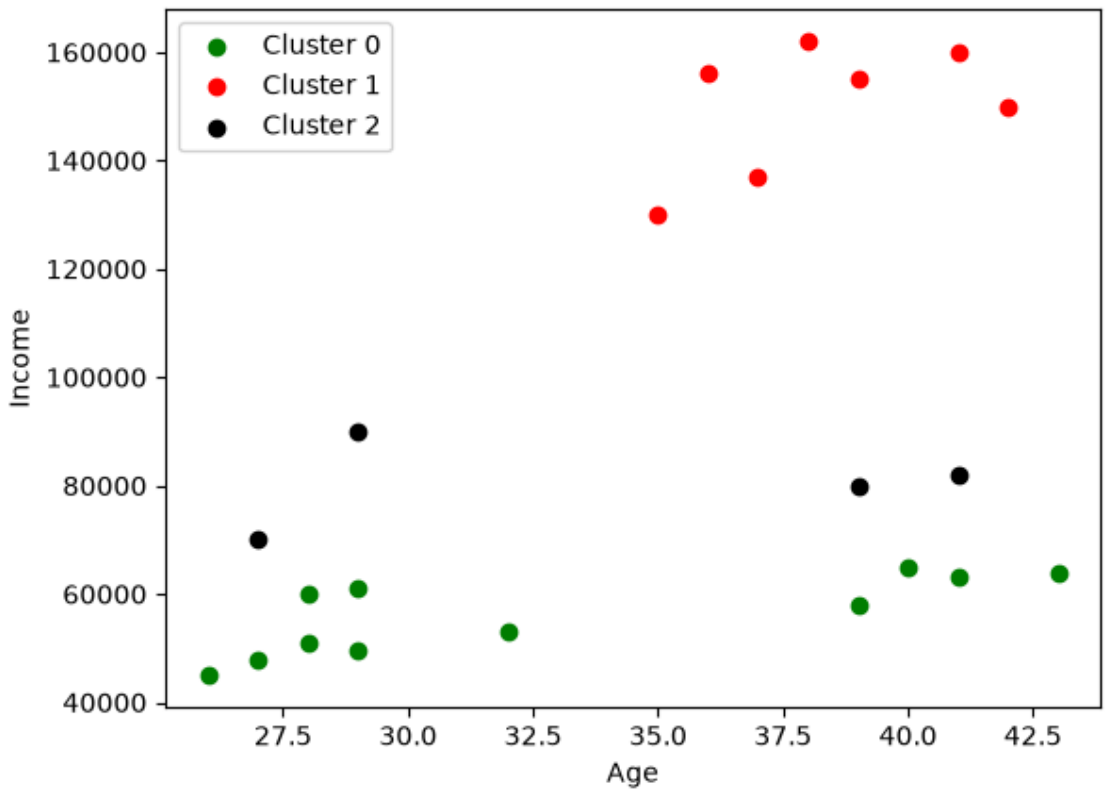
Out[6]:

| | Name | Age | Income | cluster |
|---|---|---|---|---|
| **0** | Ram | 27 | 70000 | 2 |
| **1** | Shyam | 29 | 90000 | 2 |
| **2** | Manish | 29 | 61000 | 0 |
| **3** | Arjun | 28 | 60000 | 0 |
| **4** | Abhishek | 42 | 150000 | 1 |

In [7]:
```
df1= df[df.cluster ==0]
df2= df[df.cluster ==1]
df3= df[df.cluster ==2]

plt.scatter(df1.Age,df1.Income,color='green')
plt.scatter(df2.Age,df2.Income,color='red')
plt.scatter(df3.Age,df3.Income,color='black')

plt.xlabel('Age')
plt.ylabel('Income')
plt.legend(['Cluster 0','Cluster 1','Cluster 2'])
plt.show()
```



In [8]:
```
scaler = MinMaxScaler()
scaler.fit(df[['Income']])
```

Out[8]:

MinMaxScaler

▶ Parameters

▶ Fitted attributes

▸ 1 feature

In [9]:
```
df['Income']= scaler.transform(df[['Income']])
df
```

Out[9]:

| | Name | Age | Income | cluster |
|---|---|---|---|---|
| **0** | Ram | 27 | 0.213675 | 2 |
| **1** | Shyam | 29 | 0.384615 | 2 |
| **2** | Manish | 29 | 0.136752 | 0 |
| **3** | Arjun | 28 | 0.128205 | 0 |
| **4** | Abhishek | 42 | 0.897436 | 1 |
| **5** | Karan | 39 | 0.940171 | 1 |
| **6** | Shivendra | 41 | 0.982906 | 1 |
| **7** | Varun | 38 | 1.000000 | 1 |
| **8** | Manuj | 36 | 0.948718 | 1 |
| **9** | Govind | 35 | 0.726496 | 1 |
| **10** | Sagar | 37 | 0.786325 | 1 |
| **11** | Deepak | 26 | 0.000000 | 0 |
| **12** | Aman | 27 | 0.025641 | 0 |
| **13** | Akash | 28 | 0.051282 | 0 |
| **14** | Shivay | 29 | 0.038462 | 0 |
| **15** | Rohit | 32 | 0.068376 | 0 |
| **16** | Shushank | 40 | 0.170940 | 0 |
| **17** | Mahendra | 41 | 0.153846 | 0 |
| **18** | Ashish | 43 | 0.162393 | 0 |
| **19** | Ritesh | 39 | 0.299145 | 2 |
| **20** | Ankit | 41 | 0.316239 | 2 |
| **21** | Raja | 39 | 0.111111 | 0 |

In [10]:
```
scaler.fit(df[['Age']])
```

Out[10]:

▾ MinMaxScaler

▸ Parameters

▸ Fitted attributes

▸ 1 feature

In [11]:
```
df['Age']= scaler.transform(df[['Age']])
df
```

Out[11]:

| | Name | Age | Income | cluster |
|---|---|---|---|---|
| **0** | Ram | 0.058824 | 0.213675 | 2 |
| **1** | Shyam | 0.176471 | 0.384615 | 2 |
| **2** | Manish | 0.176471 | 0.136752 | 0 |
| **3** | Arjun | 0.117647 | 0.128205 | 0 |
| **4** | Abhishek | 0.941176 | 0.897436 | 1 |
| **5** | Karan | 0.764706 | 0.940171 | 1 |
| **6** | Shivendra | 0.882353 | 0.982906 | 1 |
| **7** | Varun | 0.705882 | 1.000000 | 1 |
| **8** | Manuj | 0.588235 | 0.948718 | 1 |
| **9** | Govind | 0.529412 | 0.726496 | 1 |
| **10** | Sagar | 0.647059 | 0.786325 | 1 |
| **11** | Deepak | 0.000000 | 0.000000 | 0 |
| **12** | Aman | 0.058824 | 0.025641 | 0 |
| **13** | Akash | 0.117647 | 0.051282 | 0 |
| **14** | Shivay | 0.176471 | 0.038462 | 0 |
| **15** | Rohit | 0.352941 | 0.068376 | 0 |
| **16** | Shushank | 0.823529 | 0.170940 | 0 |
| **17** | Mahendra | 0.882353 | 0.153846 | 0 |
| **18** | Ashish | 1.000000 | 0.162393 | 0 |
| **19** | Ritesh | 0.764706 | 0.299145 | 2 |
| **20** | Ankit | 0.882353 | 0.316239 | 2 |
| **21** | Raja | 0.764706 | 0.111111 | 0 |

In [12]:
```
m = KMeans(n_clusters=3)
y_predicted=km.fit_predict(df[['Age','Income']])
y_predicted
```

Out[12]:
```
array([0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2],
      dtype=int32)
```

In [13]:
```
df['cluster']=y_predicted
df
```

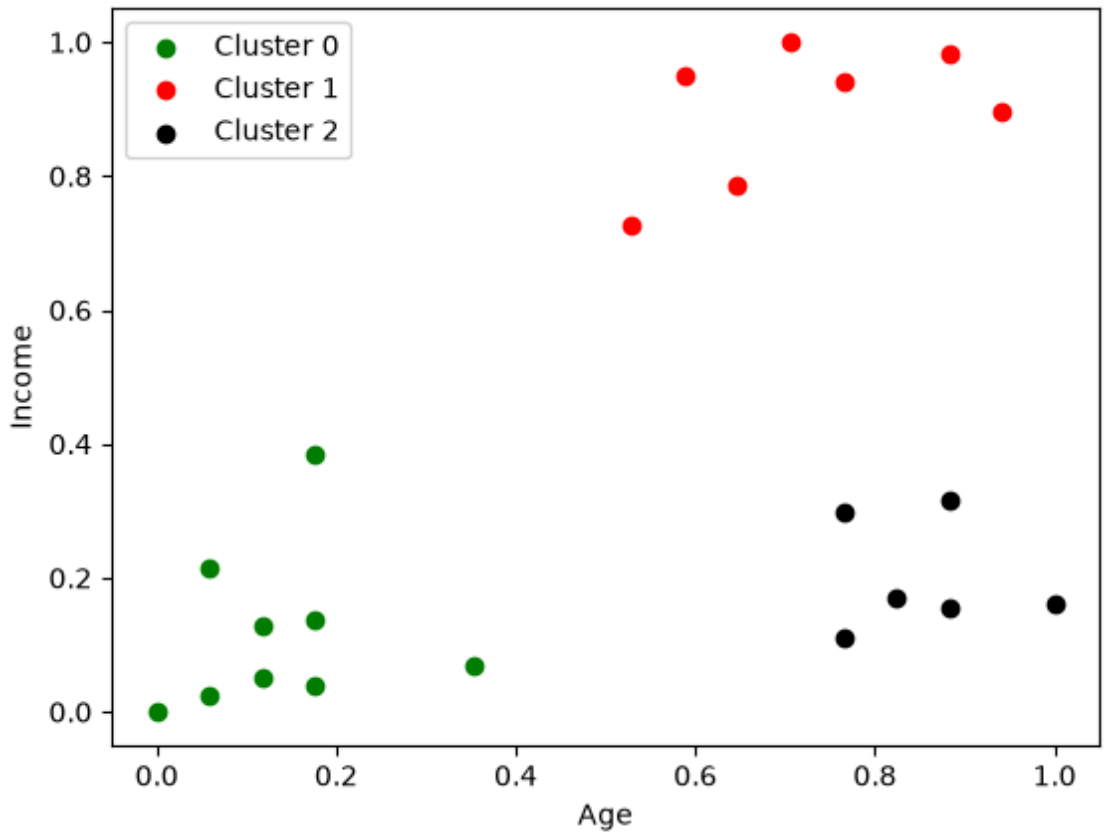
Out[13]:

| | Name | Age | Income | cluster |
|---|---|---|---|---|
| **0** | Ram | 0.058824 | 0.213675 | 0 |
| **1** | Shyam | 0.176471 | 0.384615 | 0 |
| **2** | Manish | 0.176471 | 0.136752 | 0 |
| **3** | Arjun | 0.117647 | 0.128205 | 0 |
| **4** | Abhishek | 0.941176 | 0.897436 | 1 |
| **5** | Karan | 0.764706 | 0.940171 | 1 |
| **6** | Shivendra | 0.882353 | 0.982906 | 1 |
| **7** | Varun | 0.705882 | 1.000000 | 1 |
| **8** | Manuj | 0.588235 | 0.948718 | 1 |
| **9** | Govind | 0.529412 | 0.726496 | 1 |
| **10** | Sagar | 0.647059 | 0.786325 | 1 |
| **11** | Deepak | 0.000000 | 0.000000 | 0 |
| **12** | Aman | 0.058824 | 0.025641 | 0 |
| **13** | Akash | 0.117647 | 0.051282 | 0 |
| **14** | Shivay | 0.176471 | 0.038462 | 0 |
| **15** | Rohit | 0.352941 | 0.068376 | 0 |
| **16** | Shushank | 0.823529 | 0.170940 | 2 |
| **17** | Mahendra | 0.882353 | 0.153846 | 2 |
| **18** | Ashish | 1.000000 | 0.162393 | 2 |
| **19** | Ritesh | 0.764706 | 0.299145 | 2 |
| **20** | Ankit | 0.882353 | 0.316239 | 2 |
| **21** | Raja | 0.764706 | 0.111111 | 2 |

In [14]:
```
df1= df[df.cluster ==0]
df2= df[df.cluster ==1]
df3= df[df.cluster ==2]

plt.scatter(df1.Age,df1.Income,color='green')
plt.scatter(df2.Age,df2.Income,color='red')
plt.scatter(df3.Age,df3.Income,color='black')

plt.xlabel('Age')
plt.ylabel('Income')
plt.legend(['Cluster 0','Cluster 1','Cluster 2'])
plt.show()
```



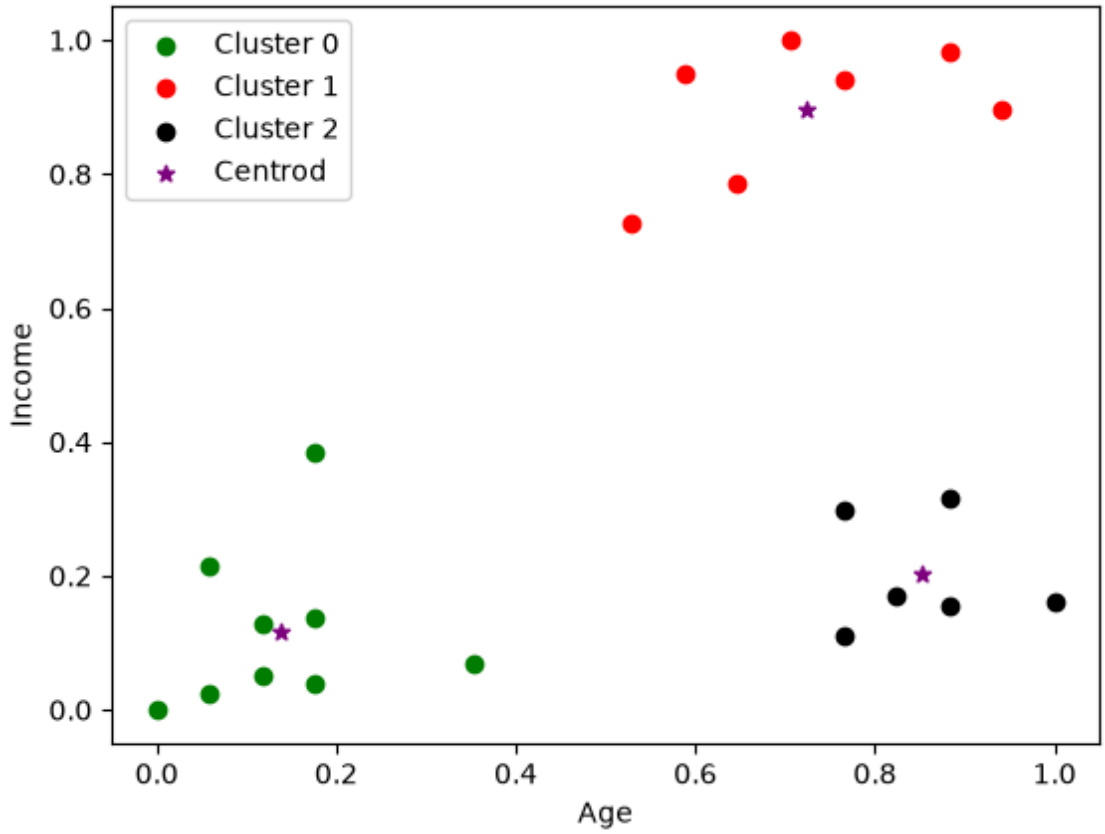

In [15]:
```
km.cluster_centers_
```

Out[15]:
```
array([[0.1372549 , 0.11633428],
       [0.72268908, 0.8974359 ],
       [0.85294118, 0.2022792 ]])
```

In [16]:
```
df1= df[df.cluster ==0]
df2= df[df.cluster ==1]
df3= df[df.cluster ==2]

plt.scatter(df1.Age,df1.Income,color='green')
plt.scatter(df2.Age,df2.Income,color='red')
plt.scatter(df3.Age,df3.Income,color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:,1],color='purple',mark

plt.xlabel('Age')
plt.ylabel('Income')
plt.legend(['Cluster 0','Cluster 1','Cluster 2','Centrod'])
plt.show()
```




In [17]:
```
k_rng=range(1,10)
sse=[]

for k in k_rng:
    km=KMeans(n_clusters=k)
    km.fit(df[['Age','Income']])
    sse.append(km.inertia_)
```

In [18]:
```
sse
```

Out[18]:
```
[5.434011511988178,
 2.0911363886990775,
 0.4750783498553096,
 0.3625079900797329,
 0.286071710668944,
 0.22624835214667535,
 0.1685851223602976,
 0.1568783380235337,
 0.11182079723210359]
```

In [19]:
```
plt.xlabel('k')
plt.ylabel('Sum of squared error')
plt.plot(k_rng,sse)
plt.show()
```

